AI Plant Disease Detection System

Project Report

1. Introduction

Agriculture plays an important role in the global economy, and plant diseases are one of the major factors affecting crop production. Early identification of diseases helps farmers take preventive measures and reduce crop losses.

The **AI Plant Disease Detection System** is a deep learning-based application developed to automatically detect plant diseases from leaf images. The system uses Convolutional Neural Networks (CNN) and Transfer Learning techniques to classify plant leaf images into different disease categories.

The application is developed using **TensorFlow, Keras, Streamlit, and Hugging Face Generative AI**. Users can upload a plant leaf image, and the system predicts the disease category along with AI-generated treatment recommendations.

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2. Objectives

The main objectives of this project are:

- To develop an AI-based plant disease detection system using deep learning.
- To classify plant leaf images into different disease categories.
- To compare the performance of CNN and transfer learning models.
- To improve classification performance using pre-trained deep learning architectures.
- To provide disease treatment recommendations using Generative AI.
- To develop an interactive user-friendly web application using Streamlit.

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3. Technologies Used

Programming Language

- Python

Deep Learning Framework

- TensorFlow
- Keras

Machine Learning Techniques

• Convolutional Neural Network (CNN)
• Transfer Learning

Deep Learning Models Used

• Custom CNN Model
• ResNet50
• EfficientNetB0
• MobileNetV2

Application Development

• Streamlit

Generative AI Integration

• Hugging Face Inference API

Data Processing Libraries

• NumPy
• Pandas
• PIL
• Matplotlib

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4. Dataset Description

The project uses a plant leaf disease image dataset containing multiple classes of healthy and diseased plant leaves.

The dataset consists of:

• Different plant varieties
• Multiple disease categories
• Healthy leaf images
• Diseased leaf images

The images are preprocessed before training. Image resizing, normalization, and data augmentation techniques are applied to improve model performance.

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5. Model Development

5.1 Baseline CNN Model

A custom Convolutional Neural Network model was developed as the baseline model.

The architecture contains:

- Convolutional Layers for feature extraction
- MaxPooling Layers for dimensionality reduction
- Dropout Layers to reduce overfitting
- Dense Layers for classification
- Softmax Output Layer for multi-class prediction

The CNN model learns important visual patterns from plant leaf images and performs disease classification.

5.2 Transfer Learning Models

Transfer learning models were implemented using pre-trained architectures.

ResNet50

ResNet50 is a deep neural network architecture that uses residual connections to solve the vanishing gradient problem. It enables efficient training of deeper networks.

EfficientNetB0

EfficientNetB0 uses optimized scaling methods to balance model accuracy and computational efficiency.

MobileNetV2

MobileNetV2 is a lightweight deep learning architecture designed for faster image classification with fewer computational resources.

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6. Model Performance Evaluation

The performance of different models was evaluated based on accuracy.

Model	Accuracy
CNN	92.90%
MobileNetV2	92.43%
ResNet50	15.55%
EfficientNetB0	2.86%

Based on the evaluation results, the custom CNN model achieved the highest accuracy and was selected as the final prediction model.

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7. Application Features

The developed Streamlit application provides the following features:

Plant Disease Prediction

• Upload a plant leaf image.

• Image preprocessing is performed automatically.

• The trained deep learning model predicts the disease category.

• The detected disease name is displayed to the user.

AI-Based Treatment Recommendation

The system provides:

• Possible treatment methods.

• Disease prevention techniques.

• AI-generated agricultural recommendations.

User Interface

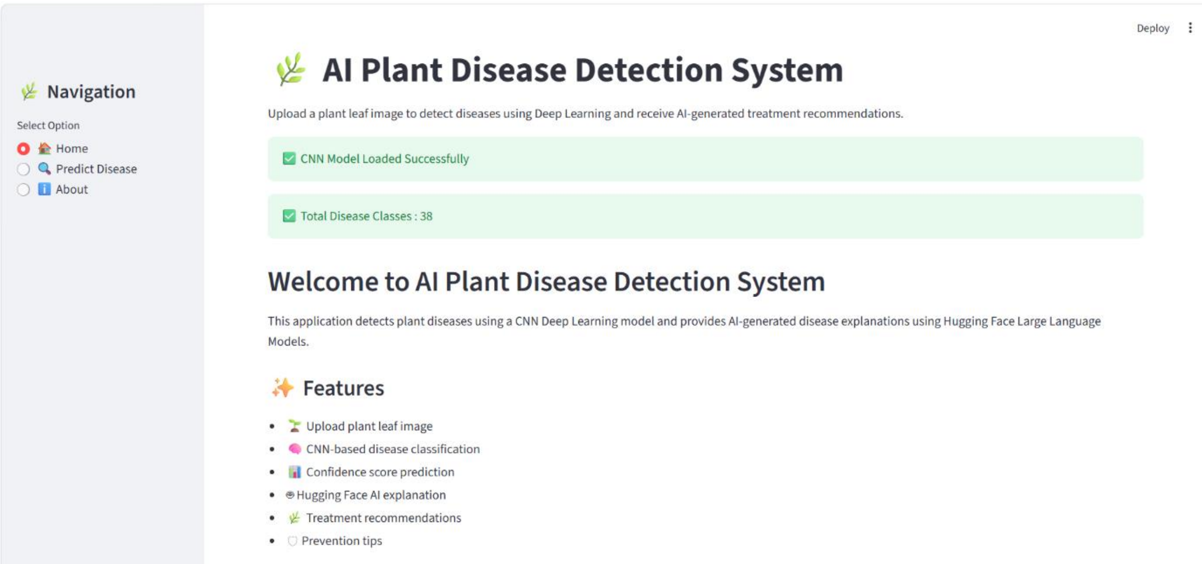
The application provides:

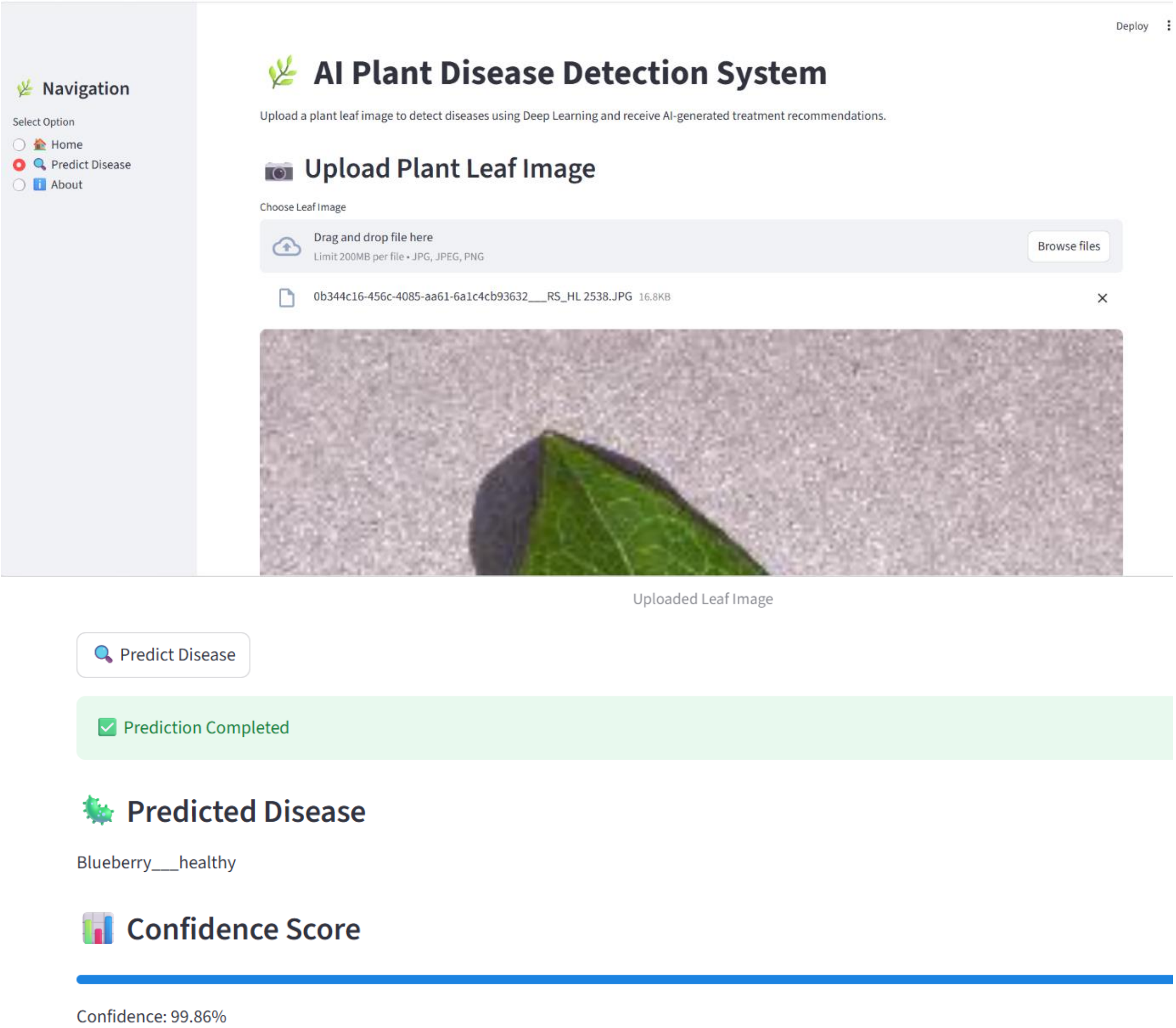
• Simple and interactive interface.

• Real-time prediction.

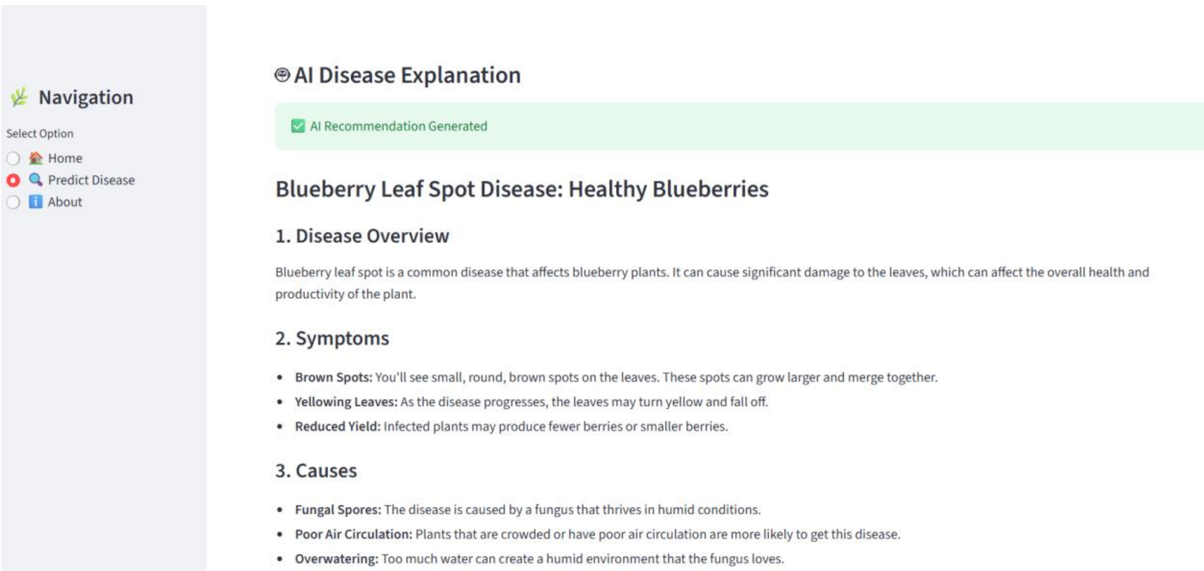
• Image display and prediction results.

Results:





Deploy
Navigation
Select Option
Home
Predict Disease
About
AI Plant Disease Detection System
Upload a plant leaf image to detect diseases using Deep Learning and receive AI-generated treatment recommendations.
Upload Plant Leaf Image
Choose Leaf Image
Drag and drop file here
Limit 200MB per file • JPG, JPEG, PNG
Browse files
0b344c16-456c-4085-aa61-6a1c4cb93632___RS_HL 2538.JPG 16.8KB
Uploaded Leaf Image
Predict Disease
Prediction Completed
Predicted Disease
Blueberry___healthy
Confidence Score
Confidence: 99.86%



Navigation
Select Option
Home
Predict Disease
About
AI Disease Explanation
AI Recommendation Generated
Blueberry Leaf Spot Disease: Healthy Blueberries
1. Disease Overview
Blueberry leaf spot is a common disease that affects blueberry plants. It can cause significant damage to the leaves, which can affect the overall health and productivity of the plant.
2. Symptoms
Brown Spots: You'll see small, round, brown spots on the leaves. These spots can grow larger and merge together.
Yellowing Leaves: As the disease progresses, the leaves may turn yellow and fall off.
Reduced Yield: Infected plants may produce fewer berries or smaller berries.
3. Causes
Fungal Spores: The disease is caused by a fungus that thrives in humid conditions.
Poor Air Circulation: Plants that are crowded or have poor air circulation are more likely to get this disease.
Overwatering: Too much water can create a humid environment that the fungus loves.

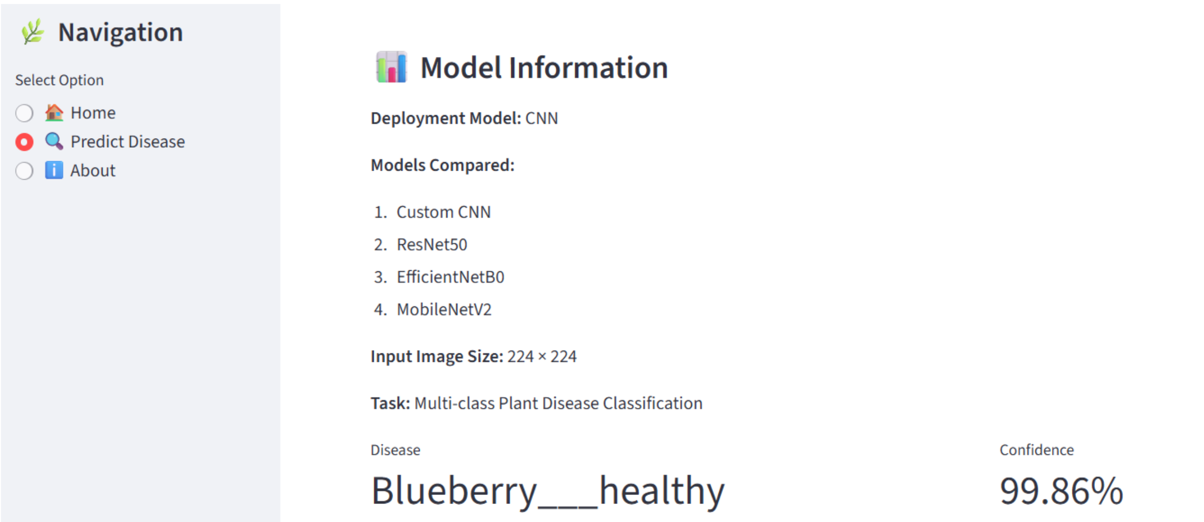
4. Recommended Treatment

- **Remove Infected Leaves:** Pick off and destroy any leaves that show signs of the disease.
- **Prune Plants:** Thin out the branches to improve air circulation and sunlight penetration.
- **Fungicides:** Use a fungicide specifically designed for blueberry leaf spot. Follow the instructions carefully.
- **Water Management:** Water the plants at the base to keep the leaves dry. Avoid overhead watering.

5. Prevention Tips

- **Choose Resistant Varieties:** Plant blueberry varieties that are resistant to leaf spot.
- **Proper Spacing:** Space your plants far enough apart to allow good air circulation.
- **Mulch:** Use a mulch around the base of the plants to help retain moisture and keep the soil cool.
- **Regular Inspection:** Check your plants regularly for any signs of the disease and act quickly if you spot any.

By following these tips, you can help keep your blueberry plants healthy and productive!



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8. Trained Model Files

Due to GitHub file size limitations, trained deep learning models are stored separately in Google Drive.

The model files include:

• Trained CNN model
• Transfer learning models ResNet50

The trained model files can be downloaded from the following Google Drive link:

Download Models from Google Drive:

https://drive.google.com/drive/folders/1zPc-vFmNNJCk-UeN3L2xAlO4ia-I5D6J?usp=sharing

After downloading, the model files should be placed inside the project models folder.

Project structure:

```
Plant-Disease-Detection/
│
├── Screenshots/
│   ├── Home.png
│   ├── Upload Leaf.png
│   ├── Prediction Result.png
│   ├── AI Recommendation.png
│   ├── AI Recommendation 2.png
│   └── Model Information.png
│
├── app.py
├── class_names.pkl
├── efficientnet_final.keras
├── mobilenet_final.keras
├── HuggingFace.ipynb
├── Model Evaluation.ipynb
├── Plant.ipynb
├── Plant_Disease_Transfer_Learning_MobileNet_ResNet.ipynb
├── model_comparison.csv
├── requirements.txt
├── .gitignore
├── Report.docx
└── README.md
```

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9. System Workflow

The working process of the system is:

1. User uploads a plant leaf image.
2. The uploaded image is resized and preprocessed.
3. The trained deep learning model extracts important features.

4. The model predicts the disease class.
5. The predicted disease name is displayed.
6. Generative AI provides treatment recommendations.

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10. Future Enhancements

The following improvements can be implemented in the future:

• Increase the dataset size for better accuracy.

• Add more plant species and disease categories.

• Deploy the system on cloud platforms.

• Develop a mobile application for farmers.

• Add multilingual support for better accessibility.

• Integrate real-time field monitoring using IoT devices.

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11. Conclusion

The AI Plant Disease Detection System successfully demonstrates the application of deep learning and artificial intelligence in agriculture.

The system can automatically identify plant diseases from leaf images and provide useful recommendations. The comparison of CNN and transfer learning models shows the effectiveness of deep learning approaches in image-based disease classification.

This project helps reduce manual disease identification efforts and supports smart agriculture by providing an AI-powered solution for early disease detection.